



Title: 960-0075 – ENGINEERING DRAWING STANDARDS	
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1. **SCOPE:** This standard applies to all engineering part and assembly drawings.
2. **PURPOSE:** The purpose of this document is to outline a set of standards by which design engineers shall create all AR engineering drawings.

3. REFERENCE DOCUMENTS:

ANSI/ASME Y14.5-2009	Dimensioning and Tolerancing
ANSI/ASME Y14.8-2009	Castings, Forgings, and Molded Parts
960-0015-05	Material Approval Process (MAP)
IPC/WHMA-A-620	Requirements and Acceptance for Cable and Wire Harness Assemblies
960-99999	THE ENGINEERING CHANGE ORDER (ECO) PROCESS
960-00300	AR Mechanical Design Best Practices
960-0075-01	ENGINEERING DRAWING STANDARDS, SOLIDWORKS TEMPLATE LAYERS

4. DEFINITIONS:

- a. AR – Amazon Robotics
- b. Engineering Drawing: An engineering drawing is a type of technical drawing created within the technical drawing discipline used to fully and clearly define requirements for engineered items.
- c. BOM: Bill of Materials. A hierarchical list of the components that make up a part or assembly of parts.
- d. PDM: Product Data Management. A PDM system for managing CAD data.
- e. BOM Balloons: Bill of Material balloons that link parts on the drawing to the PLM system BOM.
- f. PLM System: A PLM System is the Product Lifecycle Management System (Agile).
- g. CAD – Computer Aided Drafting

The following definitions are as per IPC/WHMA-A-620A Appendix A.

- h. Cable – A group of individually insulated conductors in twisted or parallel configuration under a common sheath.
- i. Cable Assembly – A cable with plugs or connectors attached.
- j. Conductor – An uninsulated wire or the conductor of an insulated wire suitable for carrying electrical current.

- k. Connector –
 - i. (1) A device used to physically and electrically join two or more conductors.
 - ii. (2) A device used to provide mechanical connect/disconnect service for electrical components.
- l. Harness – A group of wires and cables usually made with breakouts, which are tied together or pulled into a rubber or plastic sheath. A harness provides interconnection of an electric circuit.
- m. Terminal – A device designed to terminate a conductor that is to be affixed to a post, stud, chassis, another conductor, etc., to establish an electrical connection.
- n. Wire – A wire is a slender rod or filament of drawn metal.
- o. Wire Assembly – A wire with one or both ends installed into electrical terminal(s).

5. DRAWING STANDARDS:

- a. **Drawing Templates:** AR-standard SolidWorks templates have been created and are located at the following location within PDM: [C:\AR Vault\SOLIDWORKS TEMPLATES\METRIC TEMPLATES. Includes templates for creating A-Portrait, A-Landscape, B, C, D, and E size drawings.](#)
- b. **Adopted Drafting Standard:** All drawings shall comply with the standard published in: **ANSI/ASME Y14.5-2009:** Dimensioning and Tolerancing. In addition, castings, forgings, and molded parts should comply with the standards published in: **ANSI/ASME Y14.8-2009:** Castings, Forgings, and Molded Parts.
- c. **Fundamental Drawing Rules:** The following fundamental drawing rules are paraphrased from **ASME Y14.5-2009:**
 - i. All part surfaces shall have a defined dimension and tolerance. A tolerance can be specified either by a direct tolerance applied to the dimension(s), indirect tolerancing (such as a GD&T feature control frame) for basic dimensions, default tolerances specified in the title block, or general notes.
 - ii. Dimensions and tolerances should fully define the characteristics of each feature.
 - iii. All necessary dimensions for the final part should be specified on the drawing.
 - iv. Do not overdefine the part. Use reference dimensions where appropriate.
 - v. The use of reference dimensions shall be minimized. Reference dimensions are useful for communicating non-critical

dimensions to the customer, end-user, or manufacturer, such as overall dimensions.

- vi. Dimensions shall not be subject to more than one interpretation.
- vii. The drawing should fully define a part without specifying manufacturing methods. By exception only, Engineering may restrict only certain processes be used when design intent cannot be expressed by other product requirements.
- viii. Dimensions and tolerances should be arranged to optimize the readability of the drawing. The use of section views and zoomed areas to clearly define requirements is encouraged.
- ix. A 90° basic angle is inferred where no angle is specified and surfaces are shown at right angles on a 2D orthographic view.
- x. A zero basic dimension is inferred wherever axes, center planes, or surfaces appear as coincident on a drawing.

- d. **Sheet size and Multi-sheet Drawings:** A sheet size should be selected that allows for sufficient room to fully detail the part. Multiple sheets are permissible if it is not reasonable to fit all details on one sheet. The Amazon Robotics recommended sheet size is D because it prints most easily on 11x17 and provides adequate space for most parts.

e. **Title Block:**

- i. The “PART NUMBER” block must contain the official AR part number, e.g. 420-12345. This field is automatically populated from the “Number” property of the SolidWorks drawing file. This must be manually entered into the custom property window. **DO NOT OVERRIDE THE LINKED NUMBER VALUE!**
- ii. The “REV” block must contain the numerical revision associated with the PLM system revision to which the part will be released.
 - 1. Released or to be released revisions must be whole 2-digit numbers, e.g. 01, 07, or 17.
 - 2. This field is automatically populated by SolidWorks from the last revision in the revision block. Upon setting the drawing to the “Released in Agile” state in PDM, PDM will set the revision to the defined revision in the PDM database. Care must be taken to ensure the last revision block revision, the PDM revision, and the PLM system revision match. **DO NOT OVERRIDE THE LINKED REVISION VALUE!**
 - 3. All drawings/parts released in the PLM system must contain no alphabetical revision designators or version numbers. Alphabetical revision designators or version numbers are used only for internal use, preliminary quoting and apply only to unreleased iterations of drawings and parts.
- iii. The “TITLE” block must contain the title of the part. This title must be identical to the PLM system description of the part.

1. This field is automatically populated from the “Description” property of the associated SolidWorks drawing file. This must be manually entered into the custom property window. **DO NOT OVERRIDE THE LINKED TITLE VALUE!**
 2. Titles should be in ALL UPPERCASE LETTERS and should not exceed 56 characters.
 3. Titles should be succinct yet contain enough information to accurately communicate what the part is.
 4. Titles do not need to be unique across all AR part numbers, though this should be a goal. For example, there could be several “LIFTER PLATTER R” parts that are created over the life of a product. However, each iteration of the part would have a different part number.
 5. The title should describe the part using words arranged in order of significance, starting with the words that contribute the most to uniquely defining the parts (e.g. HOUSING) and ending with the words that contribute the least (e.g. H-DU).
 - Example: ASSY, TURNTABLE, H-DU
 - Example: ASSEMBLY, USER INTERFACE UI, X-DU, EU
 - Example: PLUG DOOR, RJ45, DIAG, DISCONNECT, X-DU
- iv. The “DrawnBy” block is automatically populated by PDM with the user’s full name after the first PDM check-in is complete.
- v. The “DATE CREATED” block will contain the date in MM/DD/YYYY (e.g. 12/13/2020) format of the date when the drawing was created.
1. This block is automatically populated by the “DrawnDate” property of the SolidWorks drawing file.
- vi. The “APPROX. WEIGHT (g)” block is automatically populated from the mass properties of the SolidWorks part file. **In order for this field to be accurate, the appropriate material for the part file needs to be set.**
1. For very large parts, the name and value can be changed from grams (g) to kilograms (kg)
 2. For OTS parts, override the mass properties in SolidWorks. This will provide an accurate approximation of the assembly weight
- vii. The default tolerances will change depending upon what commodity layer is chosen on the drawing (Sheet Metal, Castings, etc.), but otherwise should not be modified.

- viii. The copyright year in the drawing templates is linked to the CAD model property and part and assembly templates will be updated every year by the CAD administrator. Legacy models and drawings will need to be updated if a revision to the drawing is being done and the copyright year does not match the current year.
- ix. The remaining blocks should not be modified.

Below is an example of a title block that meets all requirements

		TITLE:		
		CLAMSHELL, TOP, LIDAR MODULE, PRO-DU		
UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS		PART NUMBER		REV
		420-12641		02
NON-BASIC DIMENSIONS: FRACTIONAL: ± 6 ANGULAR: MACH $\pm 0.1^\circ$ BEND $\pm 0.5^\circ$ OVER 0 UP TO 6: ± 0.1 OVER 120 UP TO 400: ± 0.5 OVER 6 UP TO 30: ± 0.2 OVER 400 UP TO 1000: ± 0.8 OVER 30 UP TO 120: ± 0.3 OVER 1000: ± 1.2		APPROX. WEIGHT (g)	DRAWN BY	DATE CREATED
		555	Matthew Brady	5/27/2022
		THIRD ANGLE PROJECTION 	SIZE D	SCALE: 1:1
		SHEET 1 OF 2		
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Proprietary Rights: This material and any corresponding, or associated, components, systems, or technology, are confidential and proprietary to Amazon.com Services LLC and are protected by copyright, trademark, patent, trade secret, and other proprietary rights and laws. Except as provided by a written agreement between Amazon.com Services LLC and you, this material may not be copied, reproduced, modified, published, downloaded, posted, transmitted, or distributed, in any way. Except as expressly provided in writing, Amazon.com Services LLC and its affiliates do not grant any express or implied right to you under any patents, copyrights, trademarks, or trade secret information.				

- f. **Revision Block:** All released drawings must contain a revision block. The template for the revision block is located at the following location within PDM: <C:\AR Vault\SOLIDWORKS TEMPLATES\TABLES>
- i. The revision block is typically placed in the upper right corner of the drawing.
 - ii. Revision block rows are organized such that the top row is the earliest revision. Lower rows will increment the revision number. The bottom row is the latest, to be released, revision row.
 - iii. REV. column: The bottom, latest, revision must match the revision in the title block.

- iv. ECO # column: Contains the ECO number that the revision will be released to in the PLM system. The field shall contain both the text "ECO-" and the numeric portion of the ECO, e.g. ECO-008809, not simply 008809.
- v. DESCRIPTION column: Contains concise meaningful descriptions of all changes. Each change description is followed by a zone designation when appropriate, e.g. A8, D7. For multi-sheet drawings, the zone designation includes the sheet number, e.g. SH2:A8, where "SH2" signifies sheet 2. See 960-00300 for details on best practices for formulating revision descriptions.
- vi. UPDATED column: Lists the date that the change was made in the format DD-MON-YY (e.g. 02-JUL-19). **This date should be updated each time the drawing is modified and checked into the vault.**
- vii. UPDATED BY column: Name of the person doing the new release in the format of: "First Initial". "Middle Initial". "Last Name", e.g. M. A. BRADY. It is acceptable to use a smaller font or to extend the column width to fit longer names.
- viii. Associated with the revision block, and placed on the drawing, shall be revision symbols placed next to any change. These symbols are the 2-digit revision surrounded by a hexagon, color coded such that the latest revision symbol color is in pink, e.g.  .
 - 1. Only the latest revision symbol be on the drawing. Older revision symbols will be removed.
 - 2. One revision symbol shall be placed next to the new revision in the revision history table.
- ix. Optionally, if a revision block becomes too long, earlier revision rows can be removed. At minimum, the latest revision row and previous revision row will remain in the revision block.

Below is an example of a revision block:

REVISION HISTORY				
REV.	ECO #	DESCRIPTION	UPDATED	UPDATED BY
05	ECO-014838	VARIOUS UPDATES TO FITS FROM TOLERANCE STACK UP RESULTS; RELEASE TO PILOT	09-DEC-19	M. A. BRADY
 06	ECO-015943	DFM UPDATE TO CORE PIN ON CORE SIDE (D7)	20-FEB-20	M. A. BRADY

- g. **Drawing Notes:** Drawing notes should be used to detail and clarify requirements that cannot be effectively communicated elsewhere in the drawing. 960-0075-01 details recommended sets of notes according to the type of part or assembly. **Text marked in *ITALIC* is for example and should be modified as appropriate according to properties of the part, assembly, or document. Text should not be underlined or italic on the final print.** Additionally, default GD&T surface profile tolerance should be modified as appropriate according to the necessary tolerances.
- i. Notes are typically in the upper left of the drawing.
 - ii. All notes are in uppercase letters except where the definition of the word requires lowercase, e.g. mm, Ra, RoHS, N·m.
 - iii. The MATERIAL callout shall be defined in the NOTES section of the drawing.
 - 1. Material call outs shall include a "Find Num" number in the PLM system referencing the material spec, if applicable.
 - 2. Materials determined by AR Product Governance and Compliance to require ongoing monitoring for compliance (I.E. SR319, A380) must state "*ENVIRONMENTAL COMPLIANCE MATERIAL MONITORING REQUIRED PER QMS 960-0015-09*" after the material call out. (Example: ALUMINUM ALLOY 319.1S, UNS: A03191, *ENVIRONMENTAL COMPLIANCE MATERIAL MONITORING REQUIRED PER QMS 960-0015-09*)
 - iv. The FINISH call out shall be defined in the NOTES section of the drawing.
 - 1. Finish call outs shall include a "Find Num" number in the PLM system referencing the spec, if applicable.
 - 2. The finish call out shall define the thickness and thickness tolerance for the finishing treatment, e.g. 0.02 ±0.003 mm THICK if that is not explicitly controlled in the finish specification on the BOM.
6. **DEFAULT TOLERANCES:** The default tolerances for the associated template layers can be found in 960-0075-01.
7. **QUALITY CONTROL SYMBOLS:** Design engineers shall (with consultation with manufacturing and quality engineering) apply quality control symbols to all tolerance callouts that carry special importance to the design. Definitions of when these symbols should be applied are provided in QMS document 960-0015-05 Material Approval Process (MAP).
- a. When a dimensional callout specifies multiple dimensions and tolerances, such as a hole callout, a quality control symbol can be placed to the right of individual dimensions/tolerances and will only apply to the respective adjacent dimension/tolerance.
 - b. Alternatively, a quality control symbol can be placed under a multiple dimension/tolerance callout and it will apply to the entire callout.

8. HARNESS SPECIFIC REQUIREMENTS: The Section below applies to all Cable and Harness Assembly Engineering Drawings.

- a. **Harness run list:** The harness run list table shall contain the point of origin (SOURCE), DESTINATION, SIGNAL, wire pair (PAIR), American Wire Gauge (AWG) size (GAUGE), COLOR, and NOTES. Below is an example harness run list:

AMAZON ROBOTICS WIRE RUN LIST FOR 540-00536						
SOURCE	DESTINATION	SIGNAL	PAIR	GAUGE	COLOR	NOTES
P1-1	P2-1	CAN+	A/F	24 AWG	WHITE/ORANGE	
P1-2	P2-2	CAN-	A/F		ORANGE	
P1-3	S1-5	GND1	B		WHITE/GREEN	
S1-5	S1-2	GND1	-		-	
P1-4	LED(+)	LED_VCC	C		BLUE	
P1-5	-	LED1_Q1D	C		WHITE/BLUE	
P1-6	D1 to S1-6 (A)	!PB_IN1	B		GREEN	
P1-7	-	LED1_Q1S	D		WHITE/BROWN	
P1-8	P2-7	LED1_Q4D / LED2_Q1S	D/I		BROWN	
P1-9	S1-1	ACTIVATE1_NO	E		WHITE/GREY	
P1-10	S1-3	ACTMATE1_NC	E		GREY	
P2-3	S1-5	GND2	G		WHITE/GREEN	
P2-4	LED(+)	LED_VCC	H		BLUE	
P2-5	LED(-)	LED2_Q1D	H		WHITE/BLUE	
P2-6	D2 to S1-6 (A)	!PB_IN2	G		GREEN	
P2-8	-	LED2_Q4D	I		BROWN	
P2-9	S1-1	ACTIVATE2_NO	J		WHITE/GREY	
P2-10	S1-3	ACTMATE2_NC	J		GREY	
S1-3	-	-	-		-	
S1-6	-	-	-		-	

- b. **Measurements:** All breakouts from the main harness are to be dimensioned from the outside tips of the connectors as specified in IPC/WHMA-A-620. The only exceptions are Ring terminals, hook terminals, and fork terminals which shall be dimensioned to the center point with crosshairs. Specify all lengths as whole millimeters as specified in IPC/WHMA-A-620.

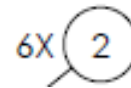
- c. **Naming Nomenclature:** Refer to document number 960-99999 for the correct part number prefix when creating new harness components. Descriptions shall be as follows:

- i. Wire description – Wire/AWG/Type (as needed)/Color i.e. Wire/8AWG/Stranded/Red
- ii. Cable description – Cable/# of conductors/type of shielding i.e. Cable/2C/FS
- iii. Connector – Conn/sex/# of row/# of positions i.e. Conn/Plug/dual row/6POS
- iv. Terminal – Term/type/AWG i.e. Term/pin/24AWG

9. Assembly Specific Requirements: The Section below applies to all Assembly Engineering Drawings.

- a. **BOM Balloons:** Each drawing shall have BOM balloons that have unique numbers for each unique part.

- i. Each BOM balloon number must match the respective “Find Num” number in the PLM system
- ii. BOM balloons are not added for documents, such as work instructions, that may be included in the PLM system BOM, unless the intent is to physically print out the document and ship it with the product.
- iii. The quantity of a particular item will be displayed to the left or right of the BOM balloon, if the quantity is greater than 1, followed by an “X”, like the following:



- iv. For clarity, it is acceptable to break out the quantity of a particular BOM balloon item as long as the sum of the quantities equals the total quantity. For example, if two sub-assemblies require 3 mounting screws each, the drawing could have two BOM balloons with 3X rather than one BOM balloon with 6X.
- v. BOM tables are not added to assembly drawings

- b. **Torques:** Screw/bolt torques are defined next to the BOM balloon of the related screw/bolt

- i. Screw/bolt torques will give the full allowable torque tolerance, typically “Nominal Torque” $\pm 10\%$

10. General Requirements: The Section below applies to all Engineering Drawings or Part files.

- a. Any UNRELEASED drawing (.sldrw), drawing export (.pdf, .dxf), CAD model (.sldprt, sldasm), CAD export (.x_t, .stp), or similar file that is to be distributed outside of PDM (ex: to vendors for quoting purposes), must conform to the following requirements:

- i. All 3D models must have "PRELIM" prepended to the file name and the version of the file in PDM appending the file name (ex: PRELIM_420-05125_V5.x_t).
- ii. All drawing file types (.PDF) must conform to the following requirements:
 1. **"NOT FOR FABRICATION"** in **bold red font**
 2. "VXX", where XX represents the version number from the PDM system
 3. Add "PRELIM" to the filename, such as PRELIM_420-05125_V07.pdf

Note: Unreleased drawings are automatically watermarked with appropriate text once uploaded to Agile.

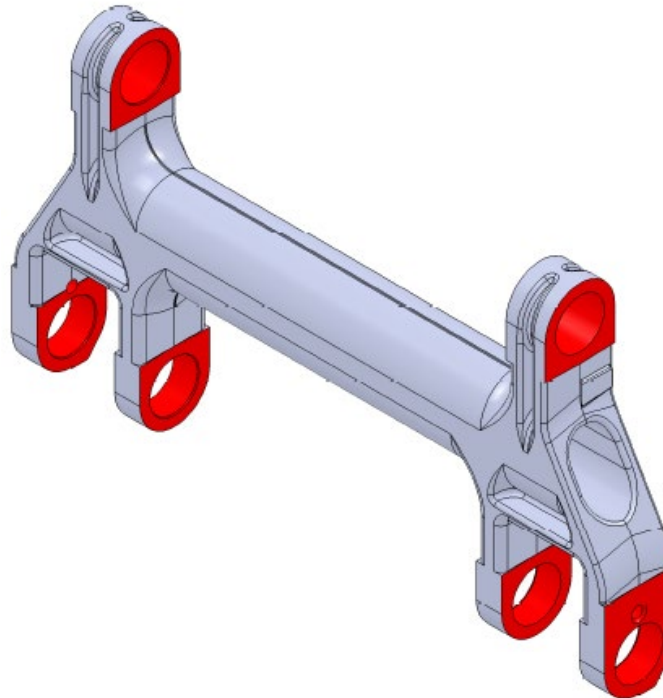
APPENDIX 1: CAST AND MOLDED PART DRAWING EXAMPLE

The following serves as an example and guideline of a cast part engineering drawing.

STEP 1: PREPARING THE 3D CAD FILE

The 3D CAD file typically serves as a part of the official documentation for cast and molded parts. A copy of the 3D CAD file should be saved with the same exact filename as the final PDF drawing (including revision), however with an extension change (.SLDPRT, .X_T, etc.). Exceptions to this include 400, 600/610, and 700/710 assemblies and corresponding drawings.

Whenever possible, the 3D CAD file should clearly communicate which surfaces are intended to be machined surfaces and which surfaces are intended to be as-cast or as-molded surfaces. This can be done by coloring all machined surfaces red (RGB 255,0,0):



STEP 2: CREATE THE DRAWING FILE

Create a drawing file using the standard sheet format, title block, and revision table in accordance with the procedures detailed earlier in this document.

STEP 3: DEFINE CRITICAL DATUMS

In general, finished cast and molded parts consist of two types of surfaces:

- Surfaces that are as-cast or as-molded
- Surfaces that are machined or otherwise formed by some secondary operation

Correspondingly, it is recommended that separate reference datum structures be specified:

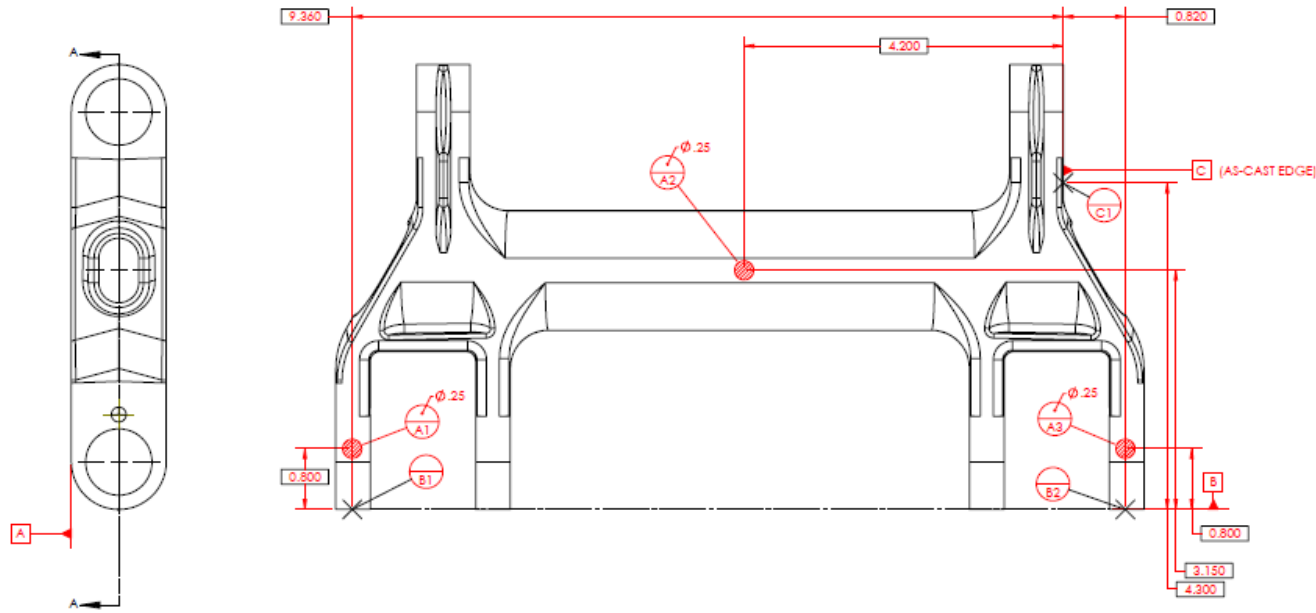
1. A datum structure based entirely upon as-cast or as-molded features
2. A datum structure based entirely upon as-machined features

When two separate datum structures are used, the features upon which one datum structure is derived must be toleranced relative to the other datum structure. This serves to limit how much all the machined surfaces can collectively “float around” as a single unit in 3D space relative to the cast geometry.

In certain cases, it may be acceptable to only specify machined datums. In such a case, the drawing notes should clearly specify a tolerance for all unspecified (cast or molded) surfaces relative to the machined datums. This tolerance must be loose enough to allow for the large uncertainties associated with cast and molded parts, yet tight enough to control how far machined features can deviate from their ideal locations on the cast geometry.

AS-CAST DATUMS

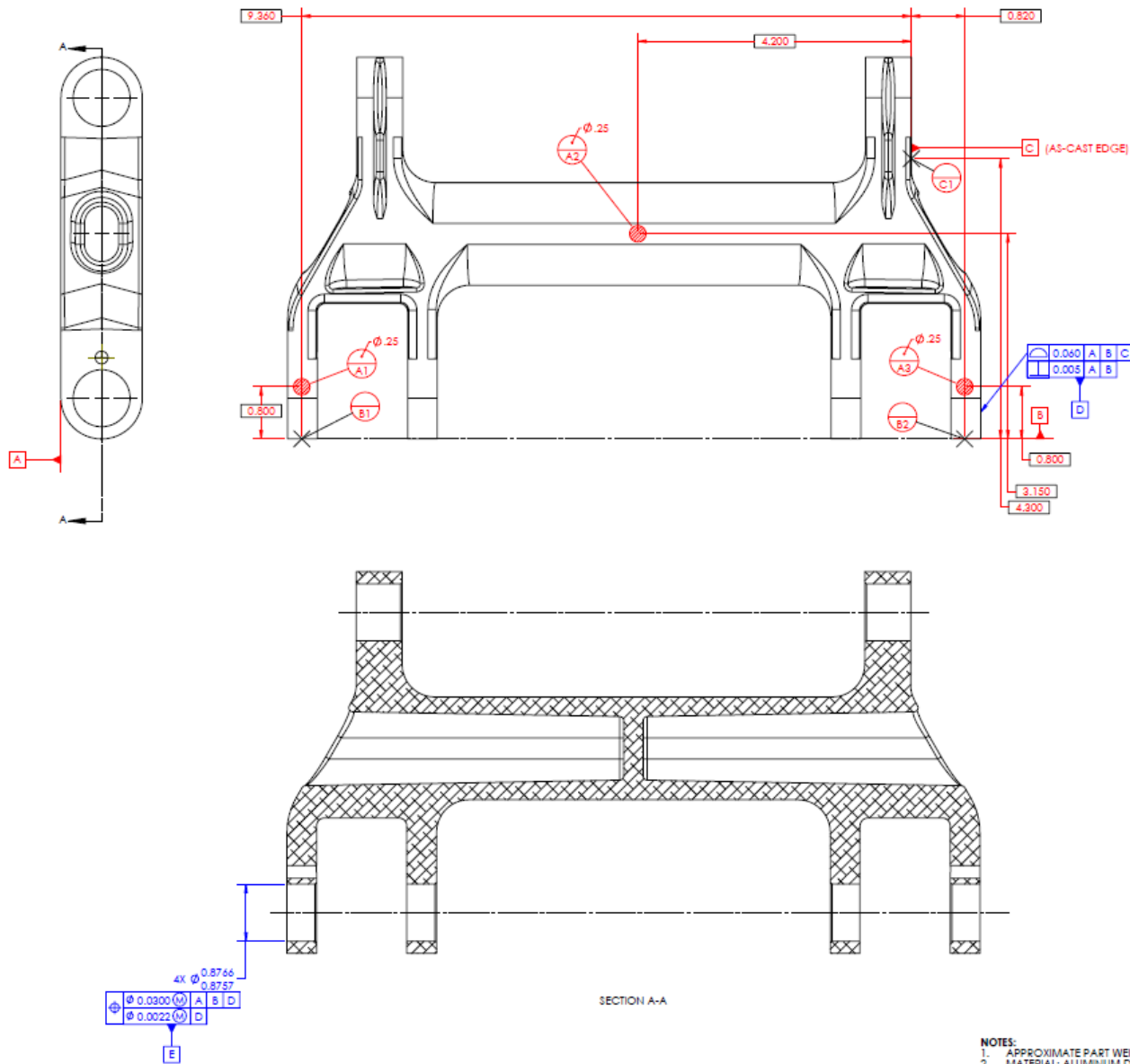
In the following example, cast features are used as datums A, B, and C. Datum A is designed to constrain the most degrees of freedom (one translational direction and two angular degrees of freedom). Datum B constrains the remaining rotational degree of freedom and another translational degree of freedom. Datum C controls the final translational degree of freedom. Dimensions, tolerances, and datum definitions that are added in this step are shown below in **RED**. **Colors are used as a communication tool for this document only and should not be used in actual engineering drawings.**



MACHINED DATUMS

The next step is to begin dimensioning and tolerancing all machined features. Generally, machined features need to have a tight tolerance relative to other machined features, but a looser tolerance relative to the cast surfaces. Therefore, it is recommended that a machined datum reference frame be established.

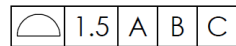
In the above example, the lower machined bore through four walls will serve as the primary machined datum. This datum allows other machined features, such as the top bores, to be dimensioned relative to the bottom bore. Since this lower machined bore constrains only four degrees of freedom, two additional datums must also be available when specifying tolerances. The second datum selected is a machined face on the far right of the part. A third datum (to control angle of the part pivoting about the primary machined datum) must also be selected. In this case, no machined features exist that are appropriate to use as a tertiary datum, so a cast datum (datum A) can serve this purpose. Dimensions, tolerances, and datum definitions that are added in this step are shown below in **BLUE**. **Colors are used as a communication tool for this document only and should not be used in actual engineering drawings.**



STEP 5: DIMENSIONING CAST AND MOLDED SURFACES

The above sections have established datum structures and fully defined the dimensions and tolerances of all machined surfaces relative to both machined datums and cast datums. However, the remaining as-cast or as-molded surfaces remain un-toleranced. For complex castings and molded parts, it is not desirable to explicitly call out a dimension and tolerance for every surface on the part. Instead, a global tolerance can be applied to all non-machined surfaces by use of a note. Further, a reference to the 3D CAD file can be given along with instructions that all geometry from the 3D CAD file should be treated as basic dimensions, e.g.:

NOTE: UNDIMENSIONED FEATURES SHOULD BE QUERIED AS BASIC FROM DIGITAL 3D DATA FILE 420-1082_R01.x_t, AND MUST BE HELD WITH THE FOLLOWING TOLERANCE:



STEP 6: POPULATE DRAWING NOTES WITH REMAINING CRITICAL INSTRUCTIONS

Drawing notes outside the standard notes should be added to the drawing as needed. Any notes with *italicized* text should be properly filled in. All text marked in *ITALIC* is for example and should be modified as appropriate according to properties of the part, assembly, or document. Text should not be underlined or italic on the final print.

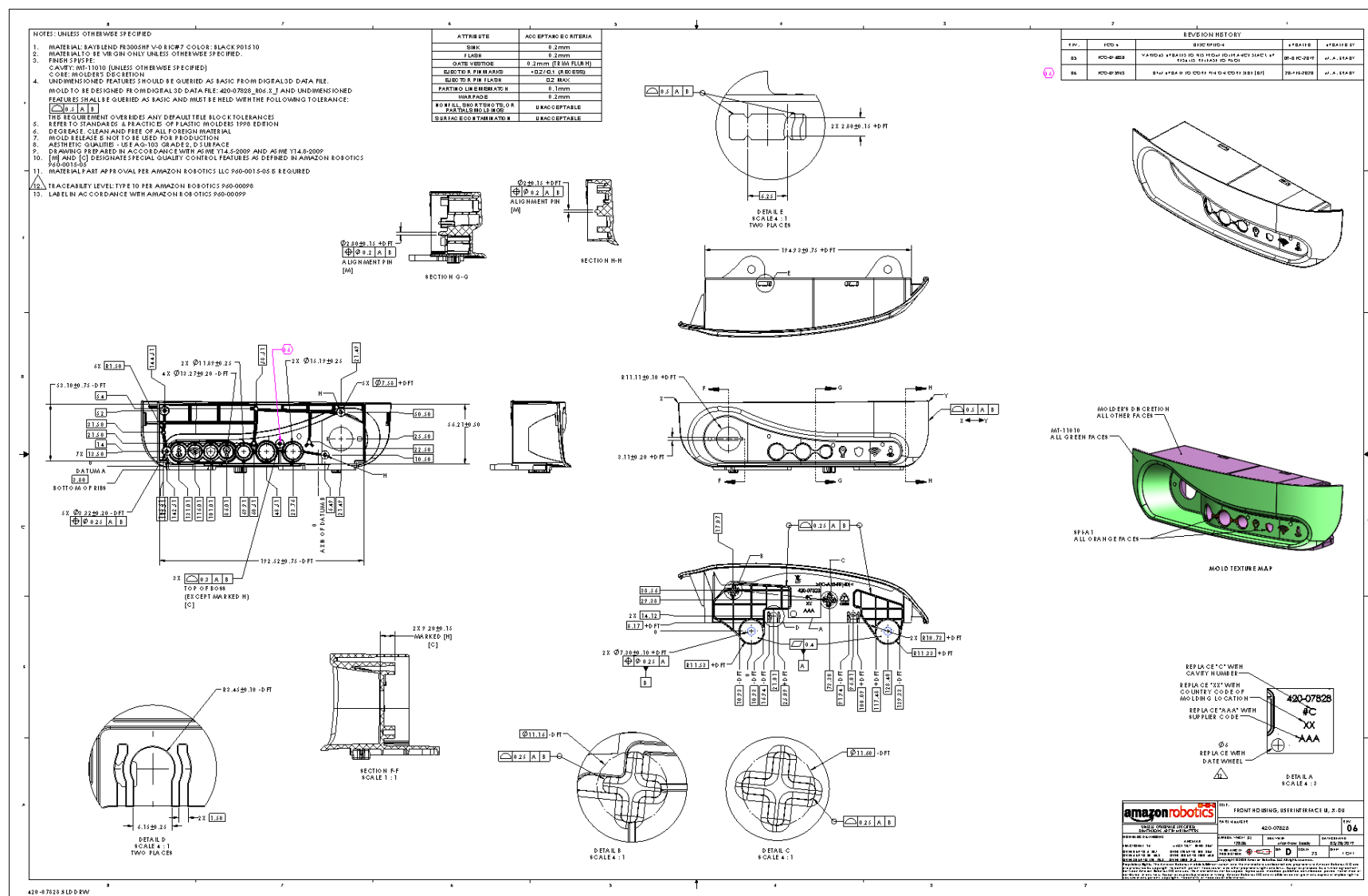
STEP 7: ADD QUALITY CONTROL SYMBOLS

If any tolerance zones meet the criteria for using quality control symbols, apply these symbols to the associated tolerance.

STEP 8: CHECK-IN AND PUBLISH DRAWING AS A PDF FILE

When a drawing is complete, the drawing should be checked into PDM and released as appropriate (set to "Pending Agile Release"). The drawing shall be published as a PDF file, taking care to ensure that the revision table and revision block are correct when publishing the file. The drawing should then be submitted with an ECO in accordance with the relevant AR procedures.

APPENDIX 2: Example drawing (see 960-00300 for further examples):



REVISION HISTORY

Rev.	ECO #	Description of Change	Sections Affected	Originator/Department
01	ECO-001609	Initial Release	All	Patrick Dingle/Hardware Engineering
02	ECO-001951	Define 'BOM Table', change Arena to Agile, change date from MM/DD/YY to YYYY-MM-DD, change title block from 'date updated' to 'date created', change RoHS from 2002/95/EC to 2011/65/EC, add 'edge break' note to all mfg processes, add safety labeling & signage to station drawings, misc minor updates	4, 5.e, 5.f, 5.g, 5.h, Appendix 2	Tom Lauducci/Hardware Engineering
03	ECO-003401	Updated Proprietary & added Safety and Regulatory, added 960-0001, added sections, removed item iv added iv, v, vi, removed header "UNOFFICIAL BOM"; added item v; Added section to define Harness specific Requirements; added Example harness Drawing	1; 3; 4.e. – n.; 5.g. iv. – vi.; 5.h.v; 6, Appendix 3	Jeremy Stockton/Tech. Ops.
04	ECO-009009	Complete update of all standard part specific drawing notes, material call outs, finish call outs for PG & C compliance.	Header, footer, all sections, Appendix 1, 2 & 3	Seth Savage – Tech Ops ME
05	ECO-010640	Added standard notes for weldments and notes for visual inspection of welds	Added Section 5 viii	Michael O'Connor/Hardware Engineering
06	ECO-011468	Added standard notes for stickers/labels with examples for vinyl labels and metal name plates, updated general requirements for unreleased documents per 960-00211	Added Section 5.g.ix and 5.g.x (pg. Error! Bookmark not defined.)	Matthew Brady/Hardware Engineering
07	ECO-015336	General updates to fix typos and make templates more relevant to today's requirements; updated various sections to match the best practices document (960-00300)	Extensive updates to various sections (3, 5, 6, 7, 8, 9, 10, Appendix 1, 2). See redlined document for details.	Matthew Brady/Hardware Engineering
08	ECO-017264	Updated cover page and replaced Amazon Robotics LLC with Amazon.com Services LLC or Amazon Robotics. Corrected spelling and added text	Updated pages 6, 9, 10, 11, and 14. Updated pages 4 section iii, page 12 note 13, page 13 note 12, page 14 note 6, page 15 note 11, page 16 note 18.	Giovanni Fraone/Hardware Engineering

		Updated note and added underline to part of note	Underlined note 1 pages 8 and 9. Updated note 2, page 8, 9, 13. Updated page 18 section b and c.	
09	ECO-017403	Remove references to the layers and block tolerances explicitly and instead leave them to be defined in 960-0075-01	5, 6	Matthew Brady/ Hardware Engineering
10	ECO-020384	Changed folder path of EPDM due to vault name change. Corrected from Linked Number to Linked Title Added note for OTS part weight Updated copyright note	3 – 5a, 6 – 5f 4 – 5e iii 5 – 5e vi 2 5 – 5e viii	Kevin Crawford/ Hardware Engineering
11	ECO-023653	Replaced vault legacy name EPDM with PDM throughout Updated Sheet Format note to Drawing Templates and added sizes Updated copyright note to clarify link to CAD model Changed Material and Finish Drawing Notes, added note to use Find Num and removed obsolete notes	3 – 5a 5 – 5e viii 8 – 5g i, ii (removed i.1, i.2, i.3, i.4, i.6, ii.1)	Kevin Crawford/ Hardware Engineering